

## Practice Midterm Exam

**Suggested Time: 75 minutes****Total: 100 points****Name:** \_\_\_\_\_**Date:** \_\_\_\_\_

**Instructions:** This practice examination is designed to simulate the format and conceptual level of the midterm examination. Try to complete it under exam conditions: closed book, closed notes, and without a calculator. Answer all questions. For multiple-choice questions, select the **single best answer**. For short-answer questions, explain your reasoning clearly and concisely.

### Part I. Multiple Choice

**40 points**

**Select the single best answer. Each question is worth 2 points.**

1. A researcher subtracts the mean of each metabolite from all measurements of that metabolite. What preprocessing operation has been performed?
  - A. Normalization
  - B. Centering
  - C. Quantile normalization
  - D. Pareto scaling
  
2. After autoscaling a variable, which statement is correct?
  - A. Its mean is 0 and its standard deviation is 1.
  - B. Its minimum is 0 and its maximum is 1.
  - C. Its distribution becomes normal.
  - D. All observations have the same value.
  
3. Two metabolites have very different variances. You want to reduce the dominance of the high-variance metabolite but preserve more of the original variance structure than autoscaling would preserve. Which method is most appropriate?
  - A. Mean centering
  - B. Pareto scaling
  - C. Quantile normalization
  - D. Log transformation
  
4. Which situation most directly motivates normalization across samples?
  - A. One variable is measured in mg/L and another in ng/L.
  - B. One variable has a much larger variance than another.

- C. Some samples have systematically higher overall signal because different amounts of material were analyzed.
  - D. A predictor is strongly correlated with the outcome.
5. What is a primary purpose of a log transformation for highly right-skewed biological measurements?
- A. Make every variable have variance 1
  - B. Reduce skewness and compress large values
  - C. Remove all outliers
  - D. Make all samples have identical distributions
6. Consider the regression model

$$\hat{Y} = 8 + 2X.$$

Which interpretation of the coefficient 2 is correct?

- A. When  $X = 0$ ,  $Y$  must equal 2.
  - B. A one-unit increase in  $X$  is associated with a 2-unit increase in predicted  $Y$ .
  - C.  $X$  explains exactly 2% of the variation in  $Y$ .
  - D. Increasing  $X$  by one unit causes  $Y$  to increase by 2 units.
7. In ordinary least squares regression, a residual is:
- A.  $\hat{y}_i - y_i$
  - B.  $y_i - \hat{y}_i$
  - C.  $y_i - \bar{y}$
  - D.  $x_i - \bar{x}$
8. Consider

$$\hat{Y} = 5 + 4X_1 - X_2.$$

What does the coefficient 4 represent?

- A.  $Y$  always increases by 4 whenever  $X_1$  increases.
  - B. Holding  $X_2$  constant, a one-unit increase in  $X_1$  is associated with a 4-unit increase in predicted  $Y$ .
  - C.  $X_1$  is four times more important than  $X_2$ .
  - D.  $X_1$  causes  $Y$  to increase by 4 units.
9. A coefficient changes substantially when a second predictor is added to a regression model. Which is the best explanation?

- A. Regression coefficients can depend on which other predictors are included in the model.
- B. Ordinary least squares no longer works when there are two predictors.
- C. Adding predictors always causes numerical instability.
- D. The original coefficient must have been calculated incorrectly.

10. Which statement about linear regression is most accurate?

- A. A statistically significant coefficient always demonstrates causation.
- B. A large coefficient always identifies the most important biological variable.
- C. Regression can describe associations, but causal conclusions require additional assumptions and study-design considerations.
- D. Confounding occurs only when the sample size is small.

11. In logistic regression, the linear predictor

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

models the:

- A. Probability directly
- B. Odds directly
- C. Log-odds
- D. Predicted class label

12. If  $P(Y = 1) = 0.5$ , the odds of  $Y = 1$  are:

- A. 0
- B. 0.5
- C. 1
- D. 2

13. In a logistic regression model, suppose

$$e^{\beta_1} = 0.5.$$

Which interpretation is correct?

- A. A one-unit increase in  $X_1$  reduces the probability by exactly 50%.
- B. A one-unit increase in  $X_1$  multiplies the odds of  $Y = 1$  by 0.5.
- C. A one-unit increase in  $X_1$  reduces the log-odds by 50%.
- D.  $P(Y = 1) = 0.5$  for every observation.

14. A classifier predicts  $P(Y = 1) = 0.42$  for a sample. Which statement is correct?
  - A. The predicted class must always be 0.
  - B. The predicted class must always be 1.
  - C. The predicted class depends on the classification threshold.
  - D. The model is incorrect because the probability is not 0 or 1.
  
15. Which metric answers the question:
 

“Of all samples predicted to be positive, what fraction are actually positive?”

  - A. Recall
  - B. Precision
  - C. Specificity
  - D. False positive rate
  
16. A medical screening test should identify as many true disease cases as possible. Which metric is especially important?
  - A. Recall
  - B. Specificity only
  - C. Training accuracy
  - D.  $R^2$
  
17. What generally happens when the classification threshold is lowered?
  - A. Fewer observations are classified as positive.
  - B. Recall generally decreases.
  - C. More observations are classified as positive.
  - D. False positives must decrease.
  
18. A classifier has  $\text{AUROC} = 0.5$ . What does this indicate?
  - A. Perfect classification
  - B. Performance equivalent to random ranking
  - C. 50% precision
  - D. 50% recall
  
19. Which pattern most strongly suggests overfitting?
  - A. Training accuracy 80%, validation accuracy 81%
  - B. Training accuracy 98%, validation accuracy 68%
  - C. Training accuracy 65%, validation accuracy 65%

D. Training accuracy 90%, validation accuracy 91%

20. A student wants to choose between  $C = 0.01$ ,  $C = 1$ , and  $C = 100$  for logistic regression. Which dataset should be used to make this hyperparameter-selection decision?

A. The final test dataset

B. Validation data or cross-validation within the development data

C. Only the training accuracy from the final fitted model

D. The complete dataset, including the test data

**Part II. Conceptual Understanding****60 points****Answer each question clearly and concisely.****21. Preprocessing Choices****10 points**

A researcher measures 500 metabolites in blood samples.

Some metabolites naturally have values around 0.01, while others have values around 10,000. In addition, one batch of samples has systematically higher overall signal intensities because more sample material was injected.

- (a) Identify the two different preprocessing problems described above. **(4 points)**
  
  
  
  
  
  
  
  
  
  
- (b) Which problem is primarily addressed by **scaling**? Explain briefly. **(2 points)**
  
  
  
  
  
  
  
  
  
  
- (c) Which problem is primarily addressed by **normalization**? Explain briefly. **(2 points)**
  
  
  
  
  
  
  
  
  
  
- (d) After autoscaling, a metabolite that originally had very little variation becomes influential in an analysis. Give one possible explanation. **(2 points)**

## 22. Interpreting a Multiple Linear Regression Model

10 points

Suppose a researcher fits the following model:

$$\widehat{\text{Blood Pressure}} = 120 + 5X_1 - 2X_2,$$

where  $X_1$  represents a measure of environmental exposure and  $X_2$  represents physical activity.

(a) Interpret the coefficient 5 for environmental exposure. **(3 points)**

(b) Interpret the coefficient  $-2$  for physical activity. **(3 points)**

(c) A researcher concludes: “Increasing environmental exposure by one unit causes blood pressure to increase by 5 units.”

Is this conclusion justified from the regression coefficient alone? Explain. **(2 points)**

(d) What does the intercept 120 represent? **(2 points)**

## 23. Understanding Confounding

8 points

A simple regression finds a strong association between coffee consumption and a health outcome.

After age is added to the model, the coefficient for coffee consumption becomes much smaller.

(a) Explain how age could act as a confounding variable in this example. **(4 points)**

(b) Why might the coefficient for coffee consumption change after age is included? **(2 points)**

(c) Does the change in the coefficient mean that the simple regression was calculated incorrectly? Explain. **(2 points)**

## 24. Logistic Regression Interpretation

8 points

Consider the logistic regression model

$$\log\left(\frac{p}{1-p}\right) = -1 + \beta_1 X.$$

Suppose

$$e^{\beta_1} = 2.$$

- (a) What does  $p$  represent in this model? (2 points)
  
- (b) Interpret  $e^{\beta_1} = 2$ . (3 points)
  
- (c) A student says: “A one-unit increase in  $X$  doubles the probability that  $Y = 1$ .”  
Explain why this statement is incorrect. (3 points)

## 25. Choosing a Classification Threshold

8 points

A classifier is used to screen patients for a serious disease. The default classification threshold is 0.5.

A physician proposes lowering the threshold to 0.2.

- (a) Will more or fewer patients generally be classified as positive? (2 points)
  
- (b) What will generally happen to recall? Explain. (2 points)
  
- (c) What will generally happen to the number of false positives? Explain. (2 points)
  
- (d) Give a reason why the physician might consider this tradeoff acceptable for screening a serious disease. (2 points)

## 26. Model Complexity and Regularization

8 points

Two logistic regression models produce the following results:

| Model | Training Accuracy | Validation Accuracy |
|-------|-------------------|---------------------|
| A     | 91%               | 88%                 |
| B     | 100%              | 71%                 |

- (a) Which model shows stronger evidence of overfitting? Explain. **(3 points)**
- (b) Based on these results, which model would you currently expect to generalize better to new data? Explain. **(2 points)**
- (c) Explain how stronger regularization might affect the coefficients and generalization performance of Model B. **(3 points)**

## 27. Cross-Validation and the Test Set

8 points

A dataset is divided into:

80% development data      and      20% test data.

The researcher wants to choose among three logistic regression hyperparameter values:

$$C = 0.01, \quad C = 1, \quad C = 100.$$

- (a) Describe how  $K$ -fold cross-validation should be used to select among these three values of  $C$ . **(3 points)**
- (b) After selecting the best value of  $C$ , what should be done before evaluating the final model on the test data? **(2 points)**
- (c) Why should the researcher not repeatedly evaluate different values of  $C$  on the test dataset and select the one with the best test performance? **(3 points)**

## Optional Self-Check Questions

**These questions are not included in the 100-point score.**

Try answering each question in one or two sentences without consulting your notes.

1. What is the conceptual difference between scaling and normalization?
2. What phrase should almost always appear when interpreting a coefficient in multiple linear regression?
3. Why is  $e^{\beta_j}$  interpreted as an odds ratio rather than a probability ratio?
4. What is the conceptual difference between precision and recall?
5. Why can accuracy be misleading when one class is rare?
6. What happens to recall when the classification threshold is lowered?
7. What pattern in training and validation performance suggests overfitting?
8. How does decreasing  $C$  affect regularization strength in scikit-learn logistic regression?
9. What is an out-of-fold prediction?
10. Why should the final test dataset remain untouched until model development is complete?

**End of Practice Examination**